

Lecture Notes: Calculus I

Complete Course: Weeks 1–10

Functions, Limits, Derivatives, Integrals, and Polar Coordinates

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Chapter 1 Functions and Sets

1.1 Sets and Their Properties

Definition 1 (Set) A set is a well-defined collection of distinct objects, considered as an object in its own right. The objects in a set are called its elements or members. If a is an element of set A , we write $a \in A$. If a is not an element of A , we write $a \notin A$. The set containing no elements is called the empty set and is denoted by $\emptyset$ or $\{\}$.

Definition 2 (Subset) If every element of set A is also an element of set B , then A is a subset of B , denoted $A \subseteq B$. If $A \subseteq B$ and $B \subseteq A$, then $A = B$. If $A \subseteq B$ but $A \neq B$, then A is a proper subset of B , denoted $A \subset B$.

Definition 3 (Set Operations) Let A and B be sets.

- The union of A and B : $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.
- The intersection of A and B : $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.
- The difference of A and B : $A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$.
- The complement of A : $\bar{A} = \{x \in U \mid x \notin A\}$ where U is the universal set.
- The Cartesian product: $A \times B = \{(a, b) \mid a \in A, b \in B\}$.

Theorem 1 (De Morgan's Laws for Sets) For any sets A and B :

$$\overline{A \cup B} = \bar{A} \cap \bar{B}, \quad \overline{A \cap B} = \bar{A} \cup \bar{B}.$$

Definition 4 (Important Number Sets) • $\mathbb{N} = \{1, 2, 3, 4, \dots\}$: natural numbers.

- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$: integers.
- $\mathbb{Q} = \{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$: rational numbers.
- $\mathbb{R}$: real numbers (rationals and irrationals).
- $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$: complex numbers.

Definition 5 (Intervals) Let $a, b \in \mathbb{R}$ with $a < b$.

- **Open**: $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.
- **Closed**: $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$.
- **Half-open**: $[a, b) = \{x \mid a \leq x < b\}$, $(a, b] = \{x \mid a < x \leq b\}$.
- **Infinite**: $(a, \infty) = \{x \mid x > a\}$, $(-\infty, b] = \{x \mid x \leq b\}$, $(-\infty, \infty) = \mathbb{R}$.

Definition 6 (Absolute Value) The absolute value of a real number x , denoted $|x|$, is

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Theorem 2 (Properties of Absolute Value) For $a, b \in \mathbb{R}$:

1. $|ab| = |a||b|$.
2. $|\frac{a}{b}| = \frac{|a|}{|b|}$ for $b \neq 0$.

3. $|a + b| \leq |a| + |b|$ (*Triangle Inequality*).
4. $|a - b| \geq ||a| - |b||$.
5. $|a| < c \iff -c < a < c$ for $c > 0$.
6. $|a| > c \iff a > c$ or $a < -c$ for $c > 0$.

1.2 Functions and Their Graphs

Definition 7 (Function) A function f from a set D to a set E is a rule that assigns to each element x in D exactly one element, called $f(x)$, in E . The set D is called the domain of f . The set E is called the codomain of f . The range (or image) of f is the set $\{f(x) \mid x \in D\}$. We write $f : D \rightarrow E$.

Definition 8 (Graph of a Function) The graph of a function f is the set of ordered pairs (x, y) such that $y = f(x)$ and x is in the domain of f .

Theorem 3 (Vertical Line Test) A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.

Definition 9 (Piecewise Defined Functions) A function may be defined by different formulas in different parts of its domain. Such a function is called piecewise defined.

Example 1 The absolute value function is a piecewise function:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

The sign function is another example:

$$\operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Definition 10 (Even and Odd Functions) Let f be a function with domain D symmetric about the origin.

- f is even if $f(-x) = f(x)$ for all $x \in D$. The graph is symmetric about the y -axis.
- f is odd if $f(-x) = -f(x)$ for all $x \in D$. The graph is symmetric about the origin.

Example 2 $f(x) = x^2$ is even. $f(x) = x^3$ is odd. $f(x) = x^2 + x$ is neither even nor odd.

Definition 11 (Increasing and Decreasing Functions) A function f is called:

- increasing on I if $x_1 < x_2 \implies f(x_1) < f(x_2)$.
- decreasing on I if $x_1 < x_2 \implies f(x_1) > f(x_2)$.
- non-decreasing on I if $x_1 < x_2 \implies f(x_1) \leq f(x_2)$.
- non-increasing on I if $x_1 < x_2 \implies f(x_1) \geq f(x_2)$.

1.3 Shifts and Scaling of Graphs

Given a basic function $y = f(x)$ and a constant $c > 0$:

- **Vertical Shift:** $y = f(x) + c$ shifts up by c . $y = f(x) - c$ shifts down by c .
- **Horizontal Shift:** $y = f(x - c)$ shifts right by c . $y = f(x + c)$ shifts left by c .
- **Vertical Scaling:** $y = cf(x)$ stretches vertically by factor c if $c > 1$, compresses if $0 < c < 1$.
- **Horizontal Scaling:** $y = f(cx)$ compresses horizontally by factor c if $c > 1$, stretches if $0 < c < 1$.
- **Reflections:** $y = -f(x)$ reflects across x -axis. $y = f(-x)$ reflects across y -axis.

Example 3 Consider $f(x) = x^2$. The graph of $y = (x - 2)^2 + 3$ is obtained by shifting the parabola right by 2 units and up by 3 units.

Example 4 The graph of $y = -2\sqrt{x+1}$ is obtained from $y = \sqrt{x}$ by: shifting left by 1, stretching vertically by factor 2, and reflecting across the x -axis.

1.4 Elementary Functions

Definition 12 (Polynomial Function) A function of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $n \geq 0$ is an integer and $a_0, a_1, \dots, a_n$ are constants called coefficients. The domain is $\mathbb{R}$.

Definition 13 (Rational Function) A function of the form $f(x) = \frac{P(x)}{Q(x)}$ where P and Q are polynomials. The domain is $\{x \mid Q(x) \neq 0\}$.

Definition 14 (Power Function) A function of the form $f(x) = x^a$ where a is a constant.

Definition 15 (Root Function) A function of the form $f(x) = \sqrt[n]{x}$.

1.5 Transcendental Functions

Definition 16 (Exponential Function) $f(x) = a^x$ where $a > 0, a \neq 1$. Domain: $\mathbb{R}$. Range: $(0, \infty)$. The most important is $f(x) = e^x$ with $e \approx 2.71828$.

Definition 17 (Logarithmic Function) $f(x) = \log_a x$ is the inverse of a^x . Domain: $(0, \infty)$. Range: $\mathbb{R}$. The natural logarithm is $\ln x = \log_e x$.

Theorem 4 (Properties of Logarithms) For $a, b > 0, a \neq 1$, and $x, y > 0$:

1. $\log_a(xy) = \log_a x + \log_a y$.
2. $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$.
3. $\log_a(x^r) = r \log_a x$.
4. $\log_a 1 = 0$ and $\log_a a = 1$.
5. $a^{\log_a x} = x$ and $\log_a(a^x) = x$.
6. Change of base: $\log_a x = \frac{\ln x}{\ln a}$.

1.6 Inverse Functions

Definition 18 (One-to-One Function) A function f is one-to-one (injective) if $f(x_1) = f(x_2) \implies x_1 = x_2$.

Theorem 5 (Horizontal Line Test) A function is one-to-one if and only if no horizontal line intersects its graph more than once.

Definition 19 (Inverse Function) Let f be one-to-one with domain D and range R . Its inverse f^{-1} has domain R and range D , defined by $f^{-1}(y) = x \iff f(x) = y$.

Theorem 6 (Properties of Inverses) 1. $f^{-1}(f(x)) = x$ for all $x \in D$.

2. $f(f^{-1}(y)) = y$ for all $y \in R$.

3. $(f^{-1})^{-1} = f$.

4. The graph of f^{-1} is the reflection of the graph of f across $y = x$.

5. If f is strictly increasing (or decreasing), then f is one-to-one.

Example 5 Find the inverse of $f(x) = 2x + 3$. Set $y = 2x + 3$, interchange: $x = 2y + 3$, solve: $y = \frac{x-3}{2}$. So $f^{-1}(x) = \frac{x-3}{2}$.

Example 6 Find the inverse of $f(x) = x^2$ for $x \geq 0$. $y = x^2$ with $x \geq 0$ implies $y \geq 0$. Interchange: $x = y^2$, solve: $y = \sqrt{x}$. So $f^{-1}(x) = \sqrt{x}$.

1.7 Composition of Functions

Definition 20 (Composition) Given f and g , the composite $(f \circ g)(x) = f(g(x))$. The domain is $\{x \in \text{dom}(g) \mid g(x) \in \text{dom}(f)\}$.

Example 7 Let $f(x) = x^2$ and $g(x) = x + 1$. Then $(f \circ g)(x) = (x + 1)^2$ and $(g \circ f)(x) = x^2 + 1$. In general, $f \circ g \neq g \circ f$.

Theorem 7 (Associativity of Composition) $f \circ (g \circ h) = (f \circ g) \circ h$.

Theorem 8 (Inverse and Composition) $f \circ f^{-1} = \text{id}_R$ and $f^{-1} \circ f = \text{id}_D$.

1.8 Trigonometric Functions

Definition 21 (Unit Circle Definition) Let t be a real number and (x, y) the terminal point of an arc of length t on the unit circle starting at $(1, 0)$. Then $\cos t = x$ and $\sin t = y$.

Definition 22 (Other Trigonometric Functions) $\tan t = \frac{\sin t}{\cos t}$, $\cot t = \frac{\cos t}{\sin t}$, $\sec t = \frac{1}{\cos t}$, $\csc t = \frac{1}{\sin t}$.

Theorem 9 (Fundamental Trigonometric Identities)

$$\sin^2 t + \cos^2 t = 1,$$

$$1 + \tan^2 t = \sec^2 t,$$

$$1 + \cot^2 t = \csc^2 t,$$

$$\sin(-t) = -\sin t \text{ (odd),}$$

$$\cos(-t) = \cos t \text{ (even).}$$

Theorem 10 (Addition and Subtraction Formulas)

$$\begin{aligned}
\sin(x + y) &= \sin x \cos y + \cos x \sin y, \\
\sin(x - y) &= \sin x \cos y - \cos x \sin y, \\
\cos(x + y) &= \cos x \cos y - \sin x \sin y, \\
\cos(x - y) &= \cos x \cos y + \sin x \sin y.
\end{aligned}$$

Theorem 11 (Double-Angle Formulas)

$$\begin{aligned}
\sin 2x &= 2 \sin x \cos x, \\
\cos 2x &= \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1.
\end{aligned}$$

Theorem 12 (Half-Angle Formulas)

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

Theorem 13 (Product-to-Sum Formulas)

$$\begin{aligned}
\sin A \cos B &= \frac{1}{2}[\sin(A - B) + \sin(A + B)], \\
\sin A \sin B &= \frac{1}{2}[\cos(A - B) - \cos(A + B)], \\
\cos A \cos B &= \frac{1}{2}[\cos(A - B) + \cos(A + B)].
\end{aligned}$$

Definition 23 (Inverse Trigonometric Functions) • $y = \arcsin x \iff \sin y = x$,
 $x \in [-1, 1]$, $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

- $y = \arccos x \iff \cos y = x$, $x \in [-1, 1]$, $y \in [0, \pi]$.
- $y = \arctan x \iff \tan y = x$, $x \in \mathbb{R}$, $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Chapter 2 Sequences and Limits**2.1 Sequences and Their Limits**

Definition 24 (Sequence) A sequence is an ordered list of numbers denoted by $\{a_n\}_{n=1}^{\infty}$ or $\{a_n\}$. Each a_n is a term. A sequence is a function with domain $\mathbb{N}$.

Example 8 Some sequences:

1. $a_n = \frac{1}{n}$: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$
2. $a_n = (-1)^n$: $-1, 1, -1, 1, \dots$
3. $a_n = \frac{n}{n+1}$: $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$
4. $a_n = 2^n$: $2, 4, 8, 16, \dots$
5. *Fibonacci*: $a_1 = 1, a_2 = 1, a_n = a_{n-1} + a_{n-2}$: $1, 1, 2, 3, 5, 8, \dots$

Definition 25 (Limit of a Sequence) A sequence $\{a_n\}$ converges to a limit L , written $\lim_{n \rightarrow \infty} a_n = L$, if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $n > N \implies |a_n - L| < \epsilon$.

Example 9 Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. Given $\epsilon > 0$, we need $|\frac{1}{n}| < \epsilon \iff n > \frac{1}{\epsilon}$. Choose $N = \lceil 1/\epsilon \rceil$. Then for $n > N$, $\frac{1}{n} < \epsilon$.

Example 10 Show that $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$. $\left| \frac{n}{n+1} - 1 \right| = \left| \frac{-1}{n+1} \right| = \frac{1}{n+1} < \epsilon \iff n+1 > \frac{1}{\epsilon}$. Choose $N = \lceil 1/\epsilon \rceil - 1$.

Definition 26 (Divergence to Infinity) $\lim_{n \rightarrow \infty} a_n = \infty$ if for every $M > 0$ there exists N such that $n > N \implies a_n > M$.

2.2 Properties of Convergent Sequences

Theorem 14 (Uniqueness of Limit) If a sequence converges, its limit is unique.

Suppose $a_n \rightarrow L_1$ and $a_n \rightarrow L_2$. Let $\epsilon = \frac{|L_1 - L_2|}{2} > 0$. There exist N_1, N_2 such that $n > N_1 \implies |a_n - L_1| < \epsilon$ and $n > N_2 \implies |a_n - L_2| < \epsilon$. Let $N = \max(N_1, N_2)$. Then for $n > N$:

$$|L_1 - L_2| \leq |L_1 - a_n| + |a_n - L_2| < 2\epsilon = |L_1 - L_2|,$$

a contradiction. Hence $L_1 = L_2$.

Theorem 15 (Algebraic Limit Theorem) If $a_n \rightarrow A$ and $b_n \rightarrow B$, then:

1. $\lim(a_n + b_n) = A + B$.
2. $\lim(a_n - b_n) = A - B$.
3. $\lim(ca_n) = cA$.
4. $\lim(a_n b_n) = AB$.
5. $\lim \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$.
6. $\lim(a_n)^p = A^p$ for $p > 0$ if $A > 0$.

Theorem 16 (Squeeze Theorem for Sequences) If $a_n \leq b_n \leq c_n$ eventually and $a_n \rightarrow L$, $c_n \rightarrow L$, then $b_n \rightarrow L$.

Example 11 Find $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$. Since $-1 \leq \sin n \leq 1$, we have $-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}$. Both bounds converge to 0. By Squeeze Theorem, the limit is 0.

2.3 Monotone Sequences

Definition 27 (Monotone Sequence) $\{a_n\}$ is increasing if $a_n \leq a_{n+1}$ for all n ; decreasing if $a_n \geq a_{n+1}$; strictly increasing if $a_n < a_{n+1}$; strictly decreasing if $a_n > a_{n+1}$. A sequence that is either increasing or decreasing is monotone.

Definition 28 (Bounded Sequence) $\{a_n\}$ is bounded above if $\exists M : a_n \leq M$ for all n . Bounded below if $\exists m : a_n \geq m$. Bounded if both.

Theorem 17 (Monotone Convergence Theorem) Every bounded monotone sequence converges. If increasing and bounded above, it converges to its supremum. If decreasing and bounded below, it converges to its infimum.

We prove the increasing case. Let $S = \{a_n : n \in \mathbb{N}\}$ and $L = \sup S$. Since $\{a_n\}$ is bounded above, L exists. Let $\epsilon > 0$. Since $L - \epsilon$ is not an upper bound, there exists N with $a_N > L - \epsilon$. Since the sequence is increasing, for $n > N$: $a_n \geq a_N > L - \epsilon$. Also $a_n \leq L$. So $|a_n - L| < \epsilon$ for $n > N$. Hence $a_n \rightarrow L$.

Example 12 $a_n = 1 - \frac{1}{n}$ is increasing and bounded above by 1, converging to 1.

Example 13 $a_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and bounded above by 3, converging to $e \approx 2.71828$.

2.4 Limits of Functions

Definition 29 (Limit of a Function) Let f be defined on an open interval containing a , except possibly at a . We say $\lim_{x \rightarrow a} f(x) = L$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$

Example 14 Show $\lim_{x \rightarrow 2} (3x + 1) = 7$. Given $\epsilon > 0$, need δ such that $0 < |x - 2| < \delta \implies |(3x + 1) - 7| < \epsilon$. This simplifies to $3|x - 2| < \epsilon$, so $|x - 2| < \epsilon/3$. Choose $\delta = \epsilon/3$.

Theorem 18 (Algebraic Limit Laws) If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:

1. $\lim(f + g) = L + M$.
2. $\lim(f - g) = L - M$.
3. $\lim(cf) = cL$.
4. $\lim(fg) = LM$.
5. $\lim \frac{f}{g} = \frac{L}{M}$ if $M \neq 0$.

Theorem 19 (One-Sided and Two-Sided Limits) $\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L$ and $\lim_{x \rightarrow a^+} f(x) = L$.

2.5 The Sandwich Theorem

Theorem 20 (Sandwich Theorem) If $f(x) \leq g(x) \leq h(x)$ near a , and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} g(x) = L$.

Let $\epsilon > 0$. There exist $\delta_1, \delta_2 > 0$ such that $0 < |x - a| < \delta_1 \implies L - \epsilon < f(x) < L + \epsilon$ and $0 < |x - a| < \delta_2 \implies L - \epsilon < h(x) < L + \epsilon$. Let $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - a| < \delta$:

$$L - \epsilon < f(x) \leq g(x) \leq h(x) < L + \epsilon,$$

so $|g(x) - L| < \epsilon$.

Example 15 $\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0$ since $-x^2 \leq x^2 \sin(1/x) \leq x^2$ and both bounds converge to 0.

Example 16 Show $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. For $0 < x < \pi/2$, geometric reasoning gives $\cos x < \frac{\sin x}{x} < 1$. Since $\lim_{x \rightarrow 0^+} \cos x = 1$, the Squeeze Theorem gives the result. Since $\frac{\sin x}{x}$ is even, the two-sided limit is 1.

Chapter 3 Limits and Continuity

3.1 One-Sided Limits

Definition 30 (Left-Hand Limit) $\lim_{x \rightarrow a^-} f(x) = L$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that $a - \delta < x < a \implies |f(x) - L| < \epsilon$.

Definition 31 (Right-Hand Limit) $\lim_{x \rightarrow a^+} f(x) = L$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that $a < x < a + \delta \implies |f(x) - L| < \epsilon$.

Example 17 For $f(x) = \frac{|x|}{x}$: $\lim_{x \rightarrow 0^+} f(x) = 1$ and $\lim_{x \rightarrow 0^-} f(x) = -1$. The two-sided limit does not exist.

3.2 Continuous Functions

Definition 32 (Continuity at a Point) f is continuous at a if:

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

Definition 33 (Continuity on an Interval) f is continuous on (a, b) if continuous at every point. On $[a, b]$ if continuous on (a, b) , right-continuous at a , and left-continuous at b .

Theorem 21 (Properties of Continuous Functions) If f and g are continuous at a and c is constant:

1. $f + g$, $f - g$, cf , fg are continuous at a .
2. $\frac{f}{g}$ is continuous if $g(a) \neq 0$.
3. If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$.

Theorem 22 (Continuity of Elementary Functions) Polynomials, rational functions, root functions, trigonometric, exponential, and logarithmic functions are continuous on their domains.

Theorem 23 (Intermediate Value Theorem) If f is continuous on $[a, b]$ and N is between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ with $f(c) = N$.

Example 18 Show $x^3 - x - 1 = 0$ has a root in $(1, 2)$. Let $f(x) = x^3 - x - 1$. $f(1) = -1 < 0$, $f(2) = 5 > 0$. By IVT, there exists $c \in (1, 2)$ with $f(c) = 0$.

Example 19 Show $\cos x = x$ has a solution. Let $f(x) = \cos x - x$. $f(0) = 1 > 0$, $f(\pi/2) = -\pi/2 < 0$. By IVT, there exists $c \in (0, \pi/2)$ with $f(c) = 0$.

3.3 Types of Discontinuities

Definition 34 (Removable Discontinuity) $\lim_{x \rightarrow a} f(x)$ exists but $f(a)$ is undefined or different. The graph has a hole.

Definition 35 (Jump Discontinuity) Both one-sided limits exist but are different.

Definition 36 (Infinite Discontinuity) At least one one-sided limit is $\pm\infty$. Vertical asymptote.

Definition 37 (Oscillating Discontinuity) Function oscillates infinitely as $x \rightarrow a$.

Example 20 $f(x) = \frac{x^2-4}{x-2}$ has removable discontinuity at $x = 2$. $f(x) = \begin{cases} x & x < 0 \\ x+1 & x \geq 0 \end{cases}$ has jump at $x = 0$. $f(x) = \frac{1}{x^2}$ has infinite at $x = 0$. $f(x) = \sin(1/x)$ has oscillating at $x = 0$.

3.4 Limits Involving Infinity

Definition 38 (Limit at Infinity) $\lim_{x \rightarrow \infty} f(x) = L$ if for every $\epsilon > 0$ there exists N such that $x > N \implies |f(x) - L| < \epsilon$.

Definition 39 (Horizontal Asymptote) $y = L$ is a horizontal asymptote if $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

Theorem 24 (End Behavior of Rational Functions) For $f(x) = \frac{P(x)}{Q(x)}$ with degrees n and m :

1. $n < m$: $\lim_{x \rightarrow \pm\infty} f = 0$.
2. $n = m$: $\lim_{x \rightarrow \pm\infty} f = \frac{a_n}{b_m}$.
3. $n > m$: limit is $\pm\infty$.

Definition 40 (Infinite Limit) $\lim_{x \rightarrow a} f(x) = \infty$ if for every $M > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta \implies f(x) > M$.

Definition 41 (Vertical Asymptote) $x = a$ is a vertical asymptote if at least one one-sided limit is $\pm\infty$.

3.5 Relative Rates of Growth

Definition 42 (Growth Rates) f grows faster than g ($f \gg g$) if $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty$. f grows slower than g ($f \ll g$) if the limit is 0. Same rate if limit is positive finite.

Theorem 25 (Hierarchy of Growth)

$$\ln x \ll x^p \ll b^x \ll x^x \quad (p > 0, b > 1).$$

Function	Equivalent	Condition
$\sin x$	x	$x \rightarrow 0$
$\tan x$	x	$x \rightarrow 0$
$\arcsin x$	x	$x \rightarrow 0$
$\arctan x$	x	$x \rightarrow 0$
$1 - \cos x$	$\frac{x^2}{2}$	$x \rightarrow 0$
$e^x - 1$	x	$x \rightarrow 0$
$\ln(1 + x)$	x	$x \rightarrow 0$
$(1 + x)^\alpha - 1$	αx	$x \rightarrow 0$

Chapter 4 Derivatives

4.1 Definition of the Derivative

Definition 43 (Derivative at a Point)

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

Definition 44 (Derivative Function)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Remark 1 (Notation) $f'(x)$, y' , $\frac{dy}{dx}$, $\frac{df}{dx}$, $D_x f(x)$.

Example 21 $f(x) = x^2$: $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = 2x$.

Example 22 $f(x) = \sqrt{x}$: $f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$.

Theorem 26 (Differentiability Implies Continuity) *If f is differentiable at a , then f is continuous at a .*

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a) = f'(a) \cdot 0 = 0.$$

Remark 2 *Converse is false: $f(x) = |x|$ is continuous at 0 but not differentiable.*

4.2 Basic Differentiation Rules

Theorem 27 (Power Rule) $\frac{d}{dx}(x^n) = nx^{n-1}$ for any real n .

Theorem 28 (Constant Multiple and Sum Rules) $(cf)' = cf'$, $(f \pm g)' = f' \pm g'$.

Theorem 29 (Product Rule) $(fg)' = f'g + fg'$.

Theorem 30 (Quotient Rule) $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$.

Example 23 $f(x) = 3x^4 - 2x^2 + 5x - 7$: $f'(x) = 12x^3 - 4x + 5$.

Example 24 $f(x) = (x^2 + 1)(x^3 - 2x)$: $f'(x) = 2x(x^3 - 2x) + (x^2 + 1)(3x^2 - 2) = 5x^4 - 3x^2 - 2$.

Example 25 $f(x) = \frac{x^2 + 1}{x - 1}$: $f'(x) = \frac{2x(x-1) - (x^2 + 1)}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$.

4.3 Derivatives of Trigonometric Functions

Theorem 31

$$\begin{aligned} (\sin x)' &= \cos x, & (\cos x)' &= -\sin x, \\ (\tan x)' &= \sec^2 x, & (\cot x)' &= -\csc^2 x, \\ (\sec x)' &= \sec x \tan x, & (\csc x)' &= -\csc x \cot x. \end{aligned}$$

For sine: $\frac{d}{dx} \sin x = \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} = \sin x \cdot 0 + \cos x \cdot 1 = \cos x$.

4.4 Derivatives of Transcendental Functions

Theorem 32

$$\begin{aligned} (e^x)' &= e^x, & (a^x)' &= a^x \ln a, \\ (\ln x)' &= \frac{1}{x}, & (\log_a x)' &= \frac{1}{x \ln a}, \\ (\arcsin x)' &= \frac{1}{\sqrt{1 - x^2}}, & (\arctan x)' &= \frac{1}{1 + x^2}. \end{aligned}$$

4.5 The Chain Rule

Theorem 33 (Chain Rule) If $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x))g'(x).$$

Example 26 $f(x) = (3x^2 + 1)^5$: $f'(x) = 5(3x^2 + 1)^4 \cdot 6x = 30x(3x^2 + 1)^4$.

Example 27 $f(x) = \sin(e^{x^2})$: $f'(x) = \cos(e^{x^2}) \cdot e^{x^2} \cdot 2x$.

4.6 Second and Higher Order Derivatives

Definition 45 $f'' = (f')'$, $f''' = (f'')'$, $f^{(n)} = (f^{(n-1)})'$.

Example 28 $f(x) = x^5 - 3x^3 + 2x$: $f' = 5x^4 - 9x^2 + 2$, $f'' = 20x^3 - 18x$, $f''' = 60x^2 - 18$, $f^{(4)} = 120x$, $f^{(5)} = 120$, $f^{(6)} = 0$.

4.7 Implicit Differentiation

Definition 46 Differentiate both sides of an equation with respect to x , treating y as a function of x , then solve for $\frac{dy}{dx}$.

Example 29 $x^2 + y^2 = 25$: $2x + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$.

Example 30 $x^3 + y^3 = 6xy$: $3x^2 + 3y^2y' = 6y + 6xy' \implies y' = \frac{2y-x^2}{y^2-2x}$.

4.8 Linearization and Differentials

Definition 47 (Linearization) $L(x) = f(a) + f'(a)(x - a)$.

Definition 48 (Differential) $dy = f'(x)dx$.

Example 31 Approximate $\sqrt{4.1}$: $f(x) = \sqrt{x}$, $a = 4$. $L(x) = 2 + \frac{1}{4}(x - 4)$. $L(4.1) = 2.025$.

4.9 Derivatives of Parametric Functions

Theorem 34 For $x = f(t)$, $y = g(t)$:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt}.$$

Example 32 $x = t^2$, $y = t^3$: $\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$, $\frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}$.

4.10 Tangents and Normal Lines

Definition 49 (Tangent Line) $y - f(a) = f'(a)(x - a)$.

Definition 50 (Normal Line) $y - f(a) = -\frac{1}{f'(a)}(x - a)$.

Example 33 $f(x) = x^2$ at $x = 1$: Tangent: $y = 2x - 1$. Normal: $y = -\frac{1}{2}x + \frac{3}{2}$.

Chapter 5 Applications of Derivatives

5.1 Maxima and Minima

Definition 51 (Absolute Extrema) $f(c)$ is absolute maximum on D if $f(c) \geq f(x)$ for all $x \in D$. Absolute minimum if $f(c) \leq f(x)$.

Definition 52 (Local Extrema) f has local maximum at c if $f(c) \geq f(x)$ for x near c . Local minimum if $f(c) \leq f(x)$.

Theorem 35 (Extreme Value Theorem) If f is continuous on $[a, b]$, it attains both absolute max and min.

Theorem 36 (Fermat's Theorem) If f has local extremum at c and $f'(c)$ exists, then $f'(c) = 0$.

Definition 53 (Critical Number) c is a critical number if $f'(c) = 0$ or $f'(c)$ does not exist.

Theorem 37 (First Derivative Test) If f' changes $+$ $\rightarrow$ $-$ at c : local max. If $-$ $\rightarrow$ $+$: local min.

Theorem 38 (Second Derivative Test) $f'(c) = 0$ and $f''(c) > 0$: local min. $f''(c) < 0$: local max.

Theorem 39 (Closed Interval Method) For absolute extrema on $[a, b]$: evaluate f at critical numbers and endpoints.

Example 34 $f(x) = x^3 - 3x^2 + 2$: $f' = 3x(x - 2)$, $f'' = 6x - 6$. Local max at $x = 0$ ($f = 2$), local min at $x = 2$ ($f = -2$).

5.2 The Mean Value Theorem

Theorem 40 (Rolle's Theorem) If f continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then exists $c \in (a, b)$ with $f'(c) = 0$.

Theorem 41 (Mean Value Theorem) If f continuous on $[a, b]$ and differentiable on (a, b) , then exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Define $g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$. Then $g(a) = g(b) = 0$. By Rolle's Theorem, exists c with $g'(c) = 0$, i.e., $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Corollary 1 If $f'(x) = 0$ on (a, b) , then f is constant. If $f'(x) > 0$, f is increasing. If $f'(x) < 0$, f is decreasing.

5.3 Graphing by First and Second Derivatives

Definition 54 (Concavity) f is concave up if f' is increasing ($f'' > 0$). Concave down if f' is decreasing ($f'' < 0$).

Definition 55 (Inflection Point) Point where concavity changes.

Theorem 42 (Second Derivative Test for Inflection) If $f''(c) = 0$ and f'' changes sign at c , then $(c, f(c))$ is an inflection point.

5.4 Asymptotes and Dominant Terms

Definition 56 (Slant Asymptote) $y = mx + b$ is a slant asymptote if $\lim_{x \rightarrow \pm\infty} [f(x) - (mx + b)] = 0$.

Theorem 43 A rational function has a slant asymptote if degree of numerator is exactly one more than denominator.

Example 35 $f(x) = \frac{x^2+1}{x-1} = x + 1 + \frac{2}{x-1}$. Slant asymptote: $y = x + 1$.

5.5 Graphing Rational Functions

Steps: factor, find domain and vertical asymptotes, find holes, find horizontal/slant asymptotes, find intercepts, analyze derivatives, sketch.

5.6 Related Rates

Theorem 44 (Strategy) Draw diagram, write equation, differentiate with respect to t , substitute known values, solve.

Example 36 Balloon inflated at 100 cm³/s. $V = \frac{4}{3}\pi r^3$, $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 5$: $100 = 100\pi \frac{dr}{dt} \implies \frac{dr}{dt} = \frac{1}{\pi}$ cm/s.

Example 37 Ladder: $x^2 + y^2 = 100$. $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$. At $x = 6$, $y = 8$, $\frac{dx}{dt} = 1$: $\frac{dy}{dt} = -\frac{3}{4}$ ft/s.

5.7 Optimization

Theorem 45 (Strategy) Identify quantity, draw diagram, introduce variables, write equation, express in one variable, find domain, find critical numbers, verify extrema.

Example 38 Two numbers summing to 20 with max product: $P = x(20 - x)$, $P' = 20 - 2x = 0 \implies x = 10$. Numbers 10 and 10.

Example 39 Box with square base, no top, volume 32. $S = x^2 + \frac{128}{x}$, $S' = 2x - \frac{128}{x^2} = 0 \implies x = 4$, $h = 2$.

5.8 Indeterminate Forms and L'Hôpital's Rule

Definition 57 (Indeterminate Forms) $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 0^0 , 1^∞ , ∞^0 .

Theorem 46 (L'Hôpital's Rule) If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

provided the latter exists.

Example 40 $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$.

Example 41 $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$.

Example 42 $\lim_{x \rightarrow 0^+} x^x$: Let $y = x^x$, $\ln y = x \ln x \rightarrow 0$. So limit is $e^0 = 1$.

Chapter 6 Integrals: Antiderivatives and Substitution

6.1 Antiderivatives

Definition 58 (Antiderivative) F is an antiderivative of f if $F'(x) = f(x)$.

Theorem 47 (General Antiderivative) If F is an antiderivative of f , the most general is $F(x) + C$.

Example 43 $f(x) = 3x^2$: $F(x) = x^3 + C$.

Example 44 $f(x) = \cos x$: $F(x) = \sin x + C$.

6.2 Indefinite Integrals

Definition 59 $\int f(x)dx = F(x) + C$ where $F' = f$.

Theorem 48 (Linearity) $\int [f + g]dx = \int fdx + \int gdx$ and $\int kfdx = k \int fdx$.

Example 45 $\int (4x^3 - 2x + 5)dx = x^4 - x^2 + 5x + C$.

6.3 Basic Integration Formulas

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \\ \int \frac{1}{x} dx &= \ln |x| + C, \\ \int e^x dx &= e^x + C, \quad \int a^x dx = \frac{a^x}{\ln a} + C, \\ \int \sin x dx &= -\cos x + C, \quad \int \cos x dx = \sin x + C, \\ \int \sec^2 x dx &= \tan x + C, \quad \int \csc^2 x dx = -\cot x + C, \\ \int \sec x \tan x dx &= \sec x + C, \quad \int \csc x \cot x dx = -\csc x + C, \\ \int \frac{1}{1+x^2} dx &= \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C.\end{aligned}$$

6.4 Substitution in Indefinite Integrals

Theorem 49 (Substitution Rule) $\int f(g(x))g'(x)dx = \int f(u)du$ where $u = g(x)$.

Example 46 $\int 2x \cos(x^2)dx = \sin(x^2) + C$.

Example 47 $\int e^{5x}dx = \frac{1}{5}e^{5x} + C$.

Example 48 $\int \frac{1}{2x+1}dx = \frac{1}{2} \ln |2x+1| + C$.

Example 49 $\int \tan x dx = -\ln |\cos x| + C = \ln |\sec x| + C$.

Theorem 50 (Trigonometric Substitution) $\sqrt{a^2 - x^2}$: $x = a \sin \theta$. $\sqrt{a^2 + x^2}$: $x = a \tan \theta$. $\sqrt{x^2 - a^2}$: $x = a \sec \theta$.

Chapter 7 Techniques of Integration

7.1 Integration by Parts

Theorem 51 (Integration by Parts)

$$\int u \, dv = uv - \int v \, du.$$

LIATE rule for choosing u : Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential.

Example 50 $\int x e^x dx = x e^x - e^x + C = e^x(x - 1) + C.$

Example 51 $\int \ln x dx = x \ln x - x + C.$

Example 52 $\int x \sin x dx = -x \cos x + \sin x + C.$

Example 53 $\int x^2 e^x dx = e^x(x^2 - 2x + 2) + C.$

Example 54 $\int e^x \sin x dx = \frac{e^x}{2}(\sin x - \cos x) + C.$

Example 55 $\int \arctan x dx = x \arctan x - \frac{1}{2} \ln(1 + x^2) + C.$

7.2 Trigonometric Integrals

Theorem 52 (Reduction Formula for $\int \sin^n x \, dx$)

$$\int \sin^n x dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x dx.$$

Theorem 53 (Reduction Formula for $\int \cos^n x \, dx$)

$$\int \cos^n x dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x dx.$$

Theorem 54 (Strategy for $\int \sin^m x \cos^n x \, dx$) If m odd, $u = \cos x$. If n odd, $u = \sin x$. If both even, use half-angle identities.

Example 56 $\int \sin^4 x dx = -\frac{1}{4} \sin^3 x \cos x + \frac{3x}{8} - \frac{3 \sin 2x}{16} + C.$

Theorem 55 (Product-to-Sum) $\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)],$ etc.

Example 57 $\int \sin 3x \cos 5x dx = \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C.$

7.3 Partial Fractions

Theorem 56 (Partial Fraction Decomposition) For proper $\frac{P(x)}{Q(x)}$:

1. Distinct linear $(ax + b)$: $\frac{A}{ax+b}.$
2. Repeated linear $(ax + b)^k$: $\sum_{j=1}^k \frac{A_j}{(ax+b)^j}.$
3. Distinct irreducible quadratic $(ax^2 + bx + c)$: $\frac{Ax+B}{ax^2+bx+c}.$
4. Repeated quadratic: sum with powers.

Example 58 $\int \frac{1}{x^2-1} dx = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C.$

Example 59 $\int \frac{2x+3}{x^2+3x+2} dx = \ln |(x+1)(x+2)| + C.$

Example 60 $\int \frac{x^2+2x-1}{x^3-x} dx = \ln \left| \frac{x(x-1)}{x+1} \right| + C.$

Example 61 $\int \frac{1}{x^2(x+1)} dx = \ln \left| \frac{x+1}{x} \right| - \frac{1}{x} + C.$

Example 62 $\int \frac{2x+1}{x^2+2x+5} dx = \ln(x^2+2x+5) - \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C.$

Chapter 8 Definite Integrals

8.1 Area and Estimating with Finite Sums

Definition 60 (Riemann Sum) $S = \sum_{i=1}^n f(c_i) \Delta x_i$ where $c_i \in [x_{i-1}, x_i]$.

Example 63 Estimate area under $f(x) = x^2$ on $[0, 1]$ with $n = 4$: right endpoints give 0.46875, left give 0.21875. Actual is $1/3$.

8.2 Sigma Notation

Theorem 57 (Properties) 1. $\sum_{i=1}^n (a_i + b_i) = \sum a_i + \sum b_i.$

2. $\sum_{i=1}^n c a_i = c \sum a_i.$

3. $\sum_{i=1}^n c = nc.$

4. $\sum_{i=1}^n i = \frac{n(n+1)}{2}.$

5. $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$

6. $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2.$

Example 64 $\int_0^1 x^2 dx = \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}.$

8.3 The Definite Integral

Definition 61 $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x.$

Theorem 58 (Properties) 1. $\int_a^a f dx = 0.$

2. $\int_a^b f dx = - \int_b^a f dx.$

3. $\int_a^b [f + g] dx = \int_a^b f dx + \int_a^b g dx.$

4. $\int_a^b c f dx = c \int_a^b f dx.$

5. $\int_a^b f dx = \int_a^c f dx + \int_c^b f dx.$

6. If $f \geq 0$ on $[a, b]$, then $\int_a^b f dx \geq 0.$

8.4 The Fundamental Theorem of Calculus

Theorem 59 (FTC Part 1) *If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t)dt$ is differentiable on (a, b) and $F'(x) = f(x)$.*

$F(x+h) - F(x) = \int_x^{x+h} f(t)dt = f(c)h$ by MVT for integrals. So $\frac{F(x+h)-F(x)}{h} = f(c)$ where c is between x and $x+h$. As $h \rightarrow 0$, $c \rightarrow x$, so $F'(x) = f(x)$.

Theorem 60 (FTC Part 2) *If f is continuous on $[a, b]$ and $F' = f$, then*

$$\int_a^b f(x)dx = F(b) - F(a).$$

Let $G(x) = \int_a^x f(t)dt$. By Part 1, $G' = f$, so $G = F + C$. Then $F(b) - F(a) = G(b) - G(a) = \int_a^b f - \int_a^a f = \int_a^b f$.

Example 65 $\int_1^3 x^2 dx = \left. \frac{x^3}{3} \right|_1^3 = \frac{26}{3}$.

Example 66 $\int_0^{\pi/2} \cos x dx = \sin x \Big|_0^{\pi/2} = 1$.

Example 67 $\int_1^e \frac{1}{x} dx = \ln x \Big|_1^e = 1$.

8.5 Substitution in Definite Integrals

Theorem 61

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Example 68 $\int_0^1 2xe^{x^2} dx = \int_0^1 e^u du = e - 1$.

Example 69 $\int_1^2 \frac{1}{2x+1} dx = \frac{1}{2} \ln \frac{5}{3}$.

Theorem 62 (Even and Odd Functions) *If f even: $\int_{-a}^a f = 2 \int_0^a f$. If f odd: $\int_{-a}^a f = 0$.*

8.6 Improper Integrals

Definition 62 (Type I) $\int_a^\infty f dx = \lim_{t \rightarrow \infty} \int_a^t f dx$.

Definition 63 (Type II) *Infinite discontinuity at endpoint or interior point.*

Theorem 63 (p -Integral Test) $\int_1^\infty \frac{1}{x^p} dx$ converges iff $p > 1$.

Example 70 $\int_1^\infty \frac{1}{x^2} dx = 1$ (converges).

Example 71 $\int_1^\infty \frac{1}{x} dx = \infty$ (diverges).

Example 72 $\int_0^1 \frac{1}{\sqrt{x}} dx = 2$ (converges).

Example 73 $\int_0^1 \frac{1}{x} dx = \infty$ (diverges).

Theorem 64 (Comparison Test) *If $0 \leq f \leq g$ on $[a, \infty)$: if $\int g$ converges then $\int f$ converges; if $\int f$ diverges then $\int g$ diverges.*

Chapter 9 Applications of Integrals

9.1 The Area Between Curves

Theorem 65 If $f(x) \geq g(x)$ on $[a, b]$:

$$A = \int_a^b [f(x) - g(x)] dx.$$

Example 74 Area between $y = x^2$ and $y = x + 2$: intersection at $x = -1, 2$. $A = \int_{-1}^2 [(x + 2) - x^2] dx = \frac{9}{2}$.

Theorem 66 With respect to y : $A = \int_c^d [R(y) - L(y)] dy$.

9.2 Volumes

Definition 64 (Disk Method) $V = \pi \int_a^b [f(x)]^2 dx$.

Definition 65 (Washer Method) $V = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx$.

Definition 66 (Cylindrical Shells) $V = 2\pi \int_a^b x f(x) dx$.

Example 75 Revolve $y = \sqrt{x}$ on $[1, 4]$ about x -axis: $V = \pi \int_1^4 x dx = \frac{15\pi}{2}$.

Example 76 Revolve region between $y = x$ and $y = x^2$ about x -axis: $V = \pi \int_0^1 (x^2 - x^4) dx = \frac{2\pi}{15}$.

Example 77 Revolve $y = x - x^2$ on $[0, 1]$ about y -axis: $V = 2\pi \int_0^1 x(x - x^2) dx = \frac{\pi}{6}$.

9.3 Arc Length

Theorem 67

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

Example 78 $y = x^{3/2}$ on $[0, 4]$: $L = \int_0^4 \sqrt{1 + \frac{9}{4}x} dx = \frac{8}{27}(10\sqrt{10} - 1)$.

Theorem 68 (Parametric) $L = \int_a^b \sqrt{(dx/dt)^2 + (dy/dt)^2} dt$.

Example 79 Circle radius r : $L = 2\pi r$.

9.4 Surface Area of Revolution

Theorem 69 (About x -axis)

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx.$$

Theorem 70 (About y -axis)

$$S = \int_c^d 2\pi g(y) \sqrt{1 + [g'(y)]^2} dy.$$

Example 80 $y = \sqrt{x}$ on $[1, 4]$: $S = \frac{\pi}{6}(17\sqrt{17} - 5\sqrt{5})$.

Example 81 Sphere radius r : $S = 4\pi r^2$.

Chapter 10 Complex Numbers and Polar Coordinates

10.1 Complex Numbers

Definition 67 $z = a + bi$, $i^2 = -1$. $\operatorname{Re}(z) = a$, $\operatorname{Im}(z) = b$.

Definition 68 (Conjugate) $\bar{z} = a - bi$.

Definition 69 (Modulus) $|z| = \sqrt{a^2 + b^2}$.

Theorem 71 (Properties) 1. $z\bar{z} = |z|^2$.

$$2. |z_1 z_2| = |z_1| |z_2|.$$

$$3. |z_1 + z_2| \leq |z_1| + |z_2|.$$

10.2 Operations with Complex Numbers

Definition 70 (Addition) $(a_1 + b_1 i) + (a_2 + b_2 i) = (a_1 + a_2) + (b_1 + b_2)i$.

Definition 71 (Multiplication) $(a_1 + b_1 i)(a_2 + b_2 i) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$.

Definition 72 (Division) $\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{|z_2|^2}$.

Example 82 $(3 + 4i)(1 - 2i) = 11 - 2i$. $\frac{3+4i}{1-2i} = -1 + 2i$.

10.3 Polar Coordinates

Theorem 72 (Conversion) $x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$, $\tan \theta = y/x$.

Example 83 $(2, \pi/6) \rightarrow (\sqrt{3}, 1)$. $(-1, 1) \rightarrow (\sqrt{2}, 3\pi/4)$.

10.4 Graphing Polar Equations

Common curves: Circle $r = a$; Cardioid $r = a(1 \pm \cos \theta)$; Rose $r = a \cos n\theta$ (n petals if odd, $2n$ if even); Lemniscate $r^2 = a^2 \cos 2\theta$.

10.5 Integration in Polar Coordinates

Theorem 73 (Area)

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta.$$

Theorem 74 (Area Between Curves)

$$A = \int_{\alpha}^{\beta} \frac{1}{2} (r_1^2 - r_2^2) d\theta.$$

Example 84 Cardioid $r = 1 + \cos \theta$: $A = \frac{3\pi}{2}$.

Example 85 One petal of $r = \cos 2\theta$: $A = \frac{\pi}{8}$.

Example 86 Inside $r = 2$, outside $r = 1 + \cos \theta$: $A = \frac{5\pi}{2}$.

10.6 Arc Length in Polar Coordinates

Theorem 75

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2} d\theta.$$

Example 87 *Cardioid* $r = 1 + \cos \theta$: $L = 8$.

Example 88 *Circle* $r = 4 \sin \theta$: $L = 4\pi$.

Example 89 *Spiral* $r = e^{\theta}$ on $[0, 2\pi]$: $L = \sqrt{2}(e^{2\pi} - 1)$.

10.7 Surface Area in Polar Coordinates

Theorem 76 *About polar axis:*

$$S = \int_{\alpha}^{\beta} 2\pi r \sin \theta \sqrt{r^2 + (dr/d\theta)^2} d\theta.$$

Example 90 *Cardioid* $r = 1 + \cos \theta$ about polar axis: $S = 4\pi$.